

Canadian Ocean Modelling workshop (29 May 2025) summary

The Canadian Ocean Modelling workshop brought together researchers to discuss ocean modelling development, biogeochemical models, and cross-institute coordination of NEMO development. Participants explored various aspects of ocean modelling, including technical challenges, potential collaborations, and the transition between different versions of the NEMO model. The workshop also included discussions on career paths in oceanography and related fields, emphasising the importance of knowledge sharing and collaboration within the modelling community.

1. Ocean model development: Globally and in Canada - discussion leaders Youyu Lu, Tahya Weiss-Gibbons and Paul Myers

The meeting focused on ocean modelling practices in Canada, with Youyu Lu presenting an overview of various ocean models and their applications. Participants discussed the challenges and benefits of transitioning between different versions of the NEMO model. The group explored the potential for greater collaboration and code sharing, particularly through platforms like GitHub. Concerns were raised about using American models due to political tensions. Discussions on vertical coordinates, forcing fields, and icebergs and which groups have tried and incorporated different methods and the need for high-resolution models along coastlines and the importance of exploring different vertical coordinate systems. The group also discussed the use of artificial intelligence for bias correction in climate models and the challenges of running high-resolution nested models for small domains. Experiences with two-way vs one-way nesting in the NEMO model for port applications were discussed. The conversation ended with a discussion on the need for a white paper to summarise the current state of ocean modelling in Canada and potential areas for future research.

2. Biogeochemical models used in Canada and future development plans- discussion leaders Susan Allen and Inge Deschepper

The second discussion session focused on biogeochemical models, where participants discussed their current work and challenges. Susan Allen highlighted the diversity of biogeochemical models compared to physical models and emphasised the need for collaboration among researchers working on similar issues. The group explored river nutrient sources, with discussions on data availability and modelling approaches, including the potential integration of chemical and temperature data from river discharge. Challenges in modelling light attenuation with regards to ice, snow, and turbidity, and carbon, sediments, and oxygen dynamics, challenges and advancements. There is a need for improved carbon uptake modelling and collaboration with chemists. Jim Christian and Amber Holdsworth discussed the limited impact of sediment models on oxygen concentrations in certain regions. Shared experiences with model resolution, bathymetry issues, and the challenges of adapting models for different environments were discussed. The group also explored the role of ballasting in particle sinking rates and the use of partial cells to reduce vertical velocity noise in models. The discussion concluded with a reminder of the trade-offs involved in choosing appropriate models for specific regions and applications.

3. Cross-institute coordination of NEMO development- discussion leaders Frédéric Dupont and Natasha Ridenour

Frédéric Dupont presented on NEMO 4 and 5 developments, highlighting Canadian contributions and potential membership in the NEMO consortium. The group discussed technical aspects of AGRIF, including its parallelisation capabilities and potential for improving model efficiency. They also addressed concerns about version compatibility between academic and government groups,

with Nicolas Lambert reporting that NEMO 5 results are similar to 4.2 when using the same parameterisations. The team agreed that NEMO 5 should be the target for future contributions, though some government groups may continue using 4.2 due to operational constraints. However, to be part of the NEMO consortium, NEMO 5 needs to be the target. Further discussions around funding, staffing, and collaboration between academia and government were held, with topics of positions and co-op opportunities. Challenges such as firewall restrictions, budget constraints, and the need for more academic involvement in government initiatives were highlighted. The conversation ended with a call for improving communication and removing barriers within the modelling community, including better use of existing resources like Slack and a shared webpage for updates and collaboration.

4. Career pathways in ocean sciences in Canada- moderated by Tahya Weiss-Gibbons and Laura Gillard

The panel discussion focused on career paths and experiences in oceanography and related fields. Martine Lezotte, Natasha Ridenour, and Roberta Hamme shared their career journeys, highlighting the importance of tailoring applications to job descriptions, particularly for government positions. They discussed the differences between academia, NGOs, and government sectors, emphasising the freedom and flexibility in academia, the collaborative environment in NGOs, and the applied focus in government roles. The panellists also addressed the challenges of graduate school, including the need for people skills and the ability to multitask, as well as the importance of networking and building a research portfolio. Further discussion focused on balancing personal and professional responsibilities in academia, particularly for women and international students. Martine shared her experiences as a parent in academia, emphasising the importance of communication with supervisors and finding supportive environments. Roberta discussed navigating the "two-body problem" in academia, highlighting the importance of flexibility and early planning. The panellists also discussed the challenges and opportunities of being an international student in academia.

5. Wrap-up- Led by Paul Myers

The conversation ended with a discussion on the value of knowledge sharing and collaboration within the NEMO community, with plans to continue these interactions and potentially produce a white paper on ocean modelling in Canada. There was also support for a new proposal to MEOPAR, once the Community of Practice call reopens, to hopefully procure three more years of funding and allow us to build on the momentum being gained this year.